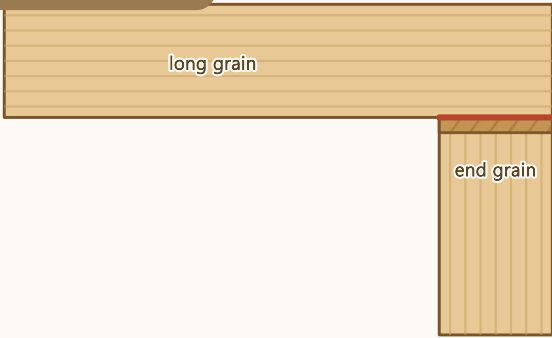


# 100 Wood Joints · the complete reference

Chinese names are the established terms where they exist ( , , , ) and a descriptive translation where there is no fixed term. The diagrams are hand-drawn technical schematics showing cut geometry — deliberately not photographs.

BUTT & REINFORCED



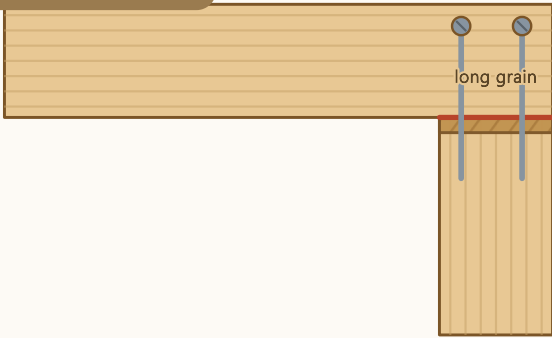
01

Butt Joint

Two boards simply placed end-to-face at 90&deg; (or end-to-end).

Easy

BUTT & REINFORCED



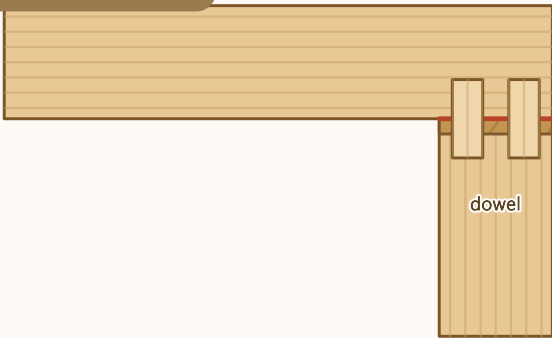
02

Screwed Butt Joint

A butt joint reinforced with screws (or nails) driven through the face of one board into the end grain of the other.

Easy

BUTT & REINFORCED



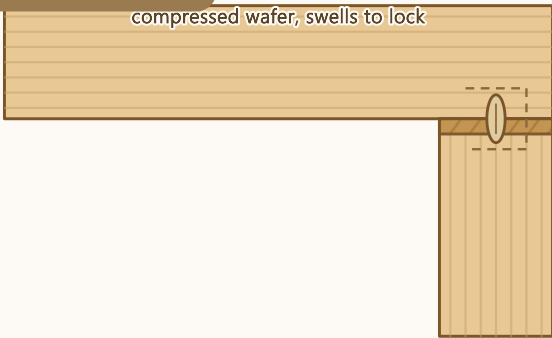
03

Dowel Joint

Holes are bored in both pieces and a fluted dowel bridges the joint.

Moderate

BUTT & REINFORCED

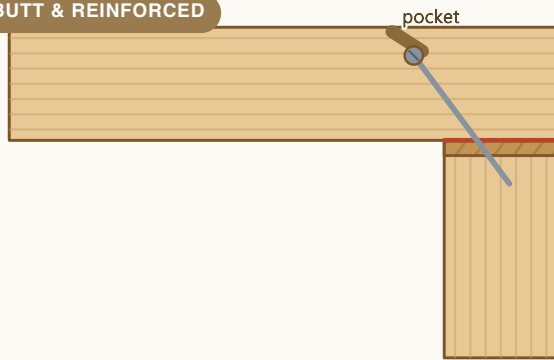


04

Biscuit Joint

A lens-shaped compressed beech wafer is glued into matching slots cut in both pieces.

Easy

**BUTT & REINFORCED**

angled screw bites long grain, not end grain

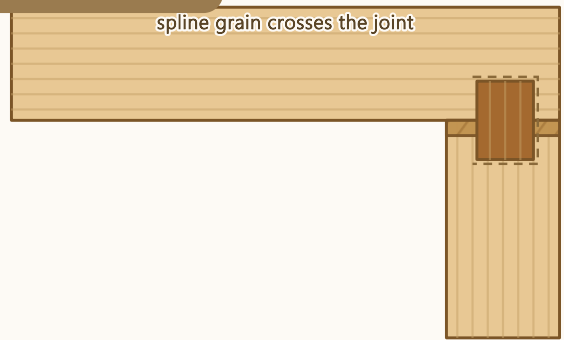
05



## Pocket-Screw Joint

An angled pilot hole (the pocket) is bored into one board, so the screw passes through it and drives into the long grain.

Easy

**BUTT & REINFORCED**

bridges two end-grain faces with long-grain glue

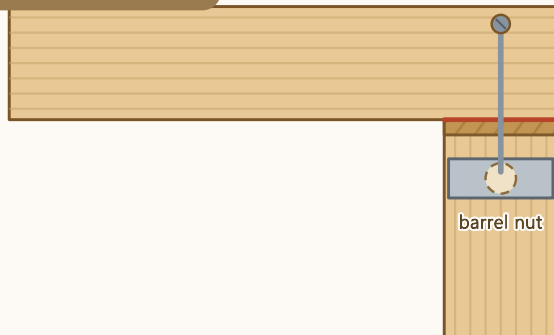
06



## Splined Butt Joint

A groove is cut in both mating faces and a separate strip — the spline — is glued in, bridging the two pieces with gen.

Moderate

**BUTT & REINFORCED**

unscrews — the knock-down joint

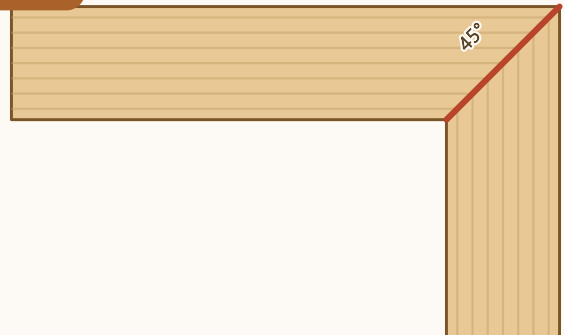
07



## Cross-Dowel (Barrel Nut) Joint

A bolt passes through one member into a barrel nut captured in a cross-drilled hole in the other.

Moderate

**MITRES**

end grain glued to end grain — the weak corner

08



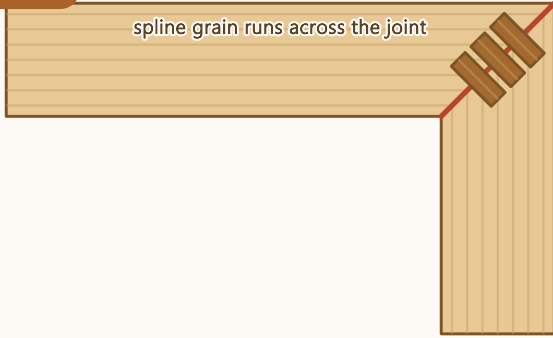
## Plain Mitre

Both boards are cut at exactly 45° so the corner closes with no end grain showing on the outside faces — the classic p.

Easy

## MITRES

spline grain runs across the joint



turns a weak mitre into a structural corner

09

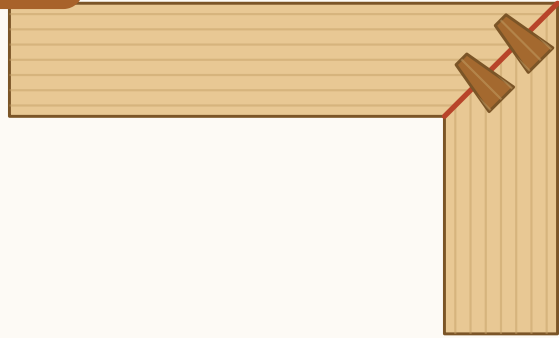


## Splined Mitre

A mitred corner with grooves cut straight across the joint from the outside.

Moderate

## MITRES



dovetail-section keys lock the mitre as well as glue it

10



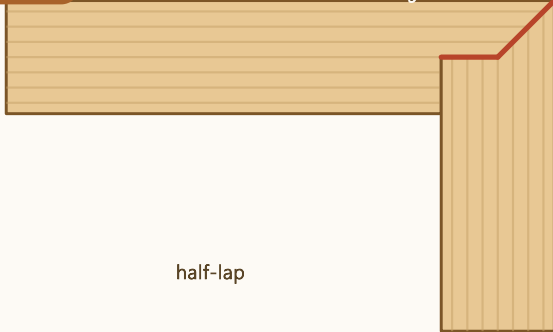
## Keyed (Feather-Key) Mitre

Instead of a straight spline, dovetail-shaped keys are let into slots cut across the mitre.

Hard

## MITRES

mitre only on the outside



half-lap

clean mitre out front, long-grain glue face behind

11

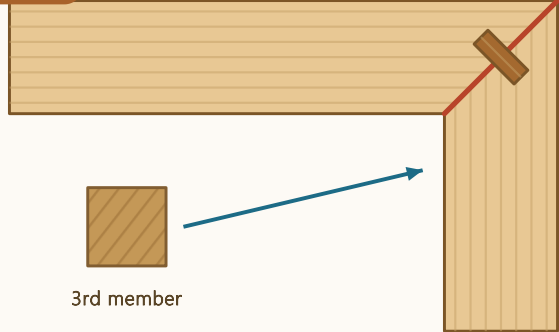


## Half-Lap Mitre (Mitred Half-Lap)

The outer corner is mitred, but the joint behind it is a half-lap: each piece loses half its thickness so they overlap.

Hard

## MITRES



3rd member

every face mitred — always reinforced inside

12

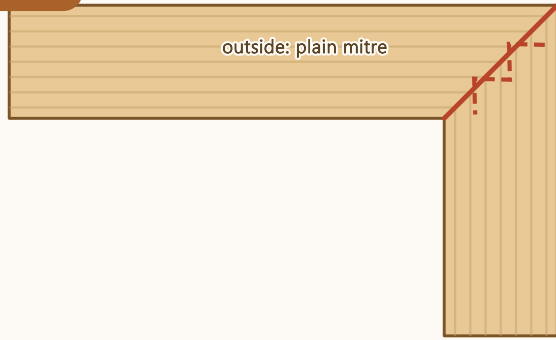


## Three-Way Mitre

Three members meeting at one corner — a leg and two rails — each mitred so the joint reads as a single clean three-way.

Expert

## MITRES



dashed = the dovetails hidden inside the corner

13

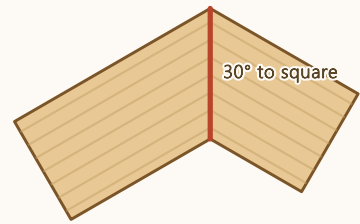


### Secret Mitred Dovetail

A dovetail is cut inside a mitred corner, so the interlock is completely hidden: the outside shows a plain mitre, and .

Expert

## MITRES



mitre angle =  $180^\circ \div \text{number of sides}$

14

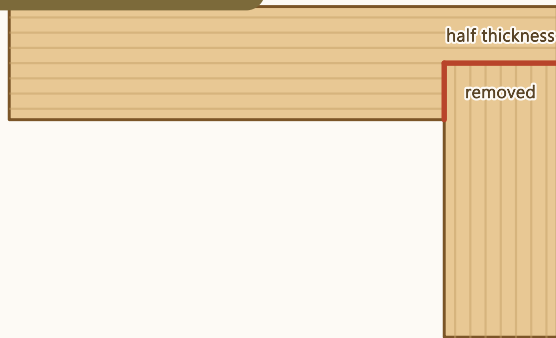


### Segmented (Polygon) Mitre

Boards mitred at an angle other than  $45^\circ$  to make a ring — hexagonal, octagonal, or the segments of a turned bowl.

Moderate

## LAPS, HALVING & BRIDLE



faces stay flush — all gluing surfaces are long grain

15

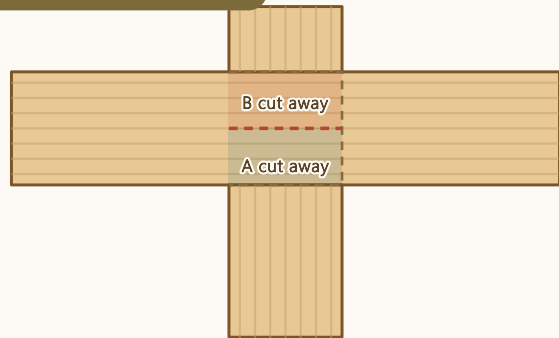


### Half-Lap Joint

Each piece has exactly half its thickness removed at the mating area, so the two overlap flush top and bottom.

Easy

## LAPS, HALVING & BRIDLE



each board loses half its thickness where they cross

16



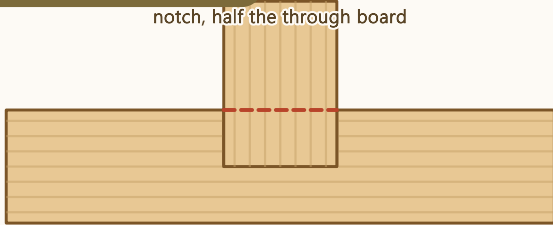
### Cross-Lap (Cross-Halving)

Two boards crossing at  $90^\circ$  in the same plane, each notched to half its thickness so they interlock flush — the classic.

Easy

#### LAPS, HALVING & BRIDLE

notch, half the through board



the notch captures the end — it cannot slide out

17



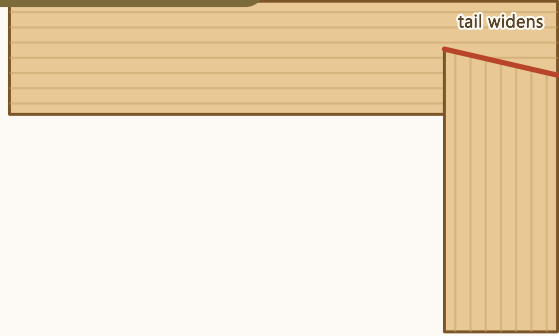
### Tee Half-Lap (T-Halving)

One board's end laps onto the middle of another — a T rather than a corner.

Easy

#### LAPS, HALVING & BRIDLE

tail widens



the taper locks the lap against pull-out

18

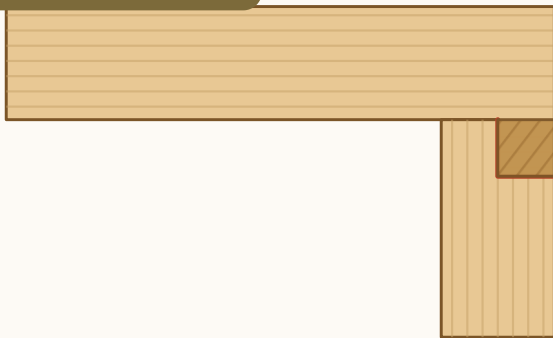


### Dovetailed Half-Lap

A half-lap whose mating faces are dovetail-shaped instead of flat.

Hard

#### LAPS, HALVING & BRIDLE



open mortise, shoulder either side — end grain shows on one face

19



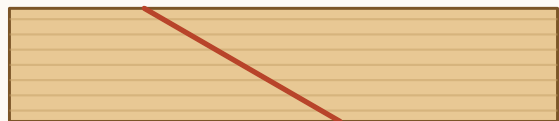
### Bridle Joint (Open Mortise & Tenon)

A lap given a proper tenon: the end of one piece is slotted into an open mortise, with a full-thickness tongue in the .

Moderate

#### LAPS, HALVING & BRIDLE

the bevel must be shallow for glue area



a scarf is measured as a ratio — 1:8 up to 1:12

20

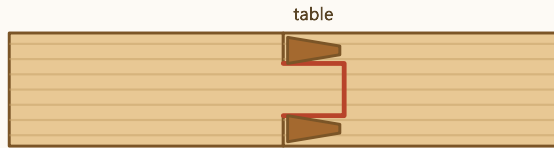


### Plain Scarf Joint

Two lengths are joined end to end on a long shallow bevel to make one continuous piece.

Moderate

#### LAPS, HALVING & BRIDLE



wedges lock it and pull the scarf up tight

21

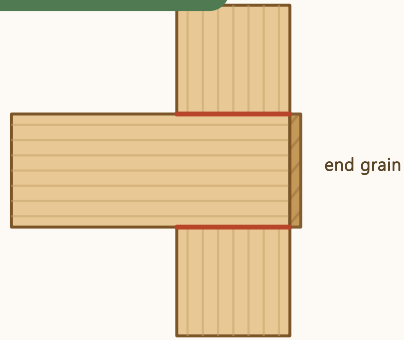


### Tabled & Wedged Scarf

A scarf with a step (the table) cut into it and a wedge driven in behind the table.

Expert

#### HOUSING & CARCASE



groove across the grain = a dado; along it = a groove

22

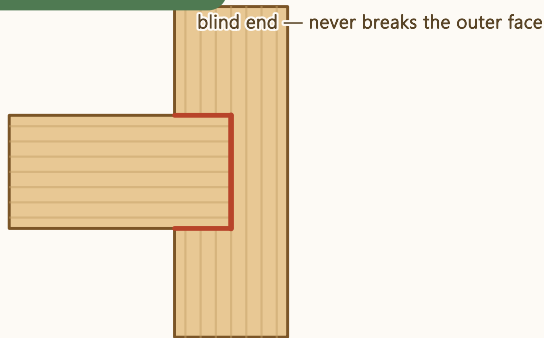


### Dado (Through Housing)

A channel cut ACROSS the grain of one board (a dado) to receive the end or edge of another.

Easy

#### HOUSING & CARCASE



the far face stays clean, so the shelf must drop in

23

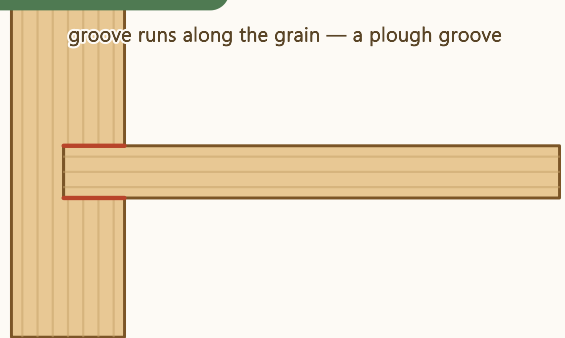


### Stopped Dado

A dado that stops before it reaches the far edge of the board, so the channel never breaks out on a visible face.

Moderate

#### HOUSING & CARCASE



a solid panel must float: never glue it along its length

24

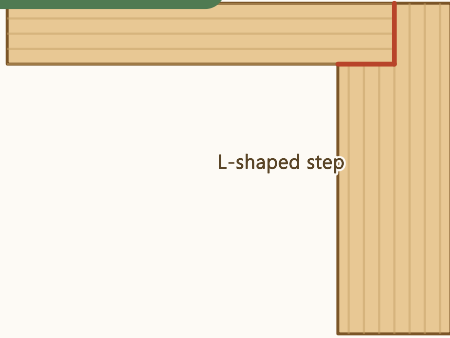


### Groove (Plough) Joint

A channel cut ALONG the grain of a board to receive a panel, a shelf, or a tongue.

Easy

#### HOUSING & CARCASE



L-shaped step

locates the pieces and adds glue face — but resists no pull-out

25

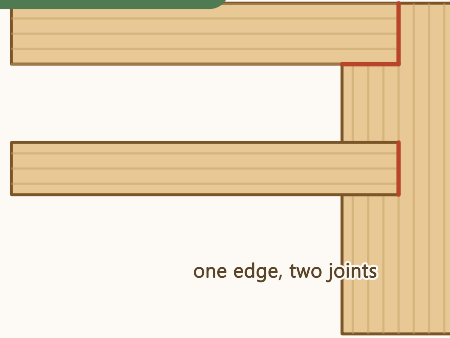


### Rabbet (Rebate) Joint

An L-shaped step cut along the edge of a board so another board can sit into it.

Easy

#### HOUSING & CARCASE



one edge, two joints

lay both out together or the corner goes weak

26

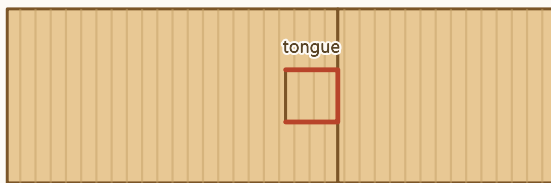


### Rabbet-and-Groove Corner

The standard cabinet corner: the side gets a rebate to take the back panel AND a groove to take a shelf or horizontal .

Moderate

#### HOUSING & CARCASE



tongue

the running tongue keeps the next board flush

27

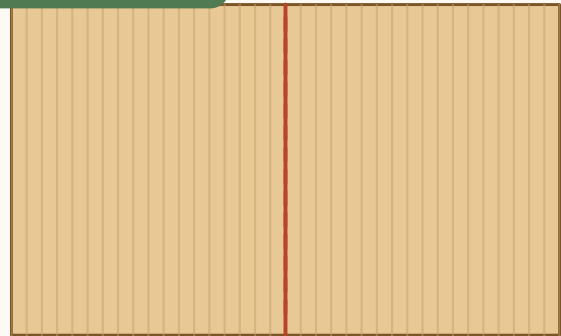


### Tongue & Groove

A projecting tongue on the edge of one board fits a matching groove in the next.

Moderate

#### HOUSING & CARCASE



long grain to long grain — stronger than the wood itself

28

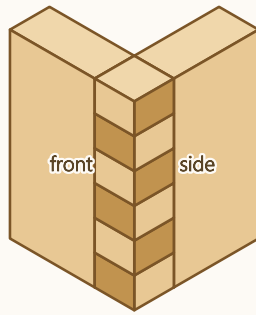


### Edge Joint (Glued Panel)

Boards joined edge to edge to make a wider panel.

Easy

#### FINGER & BOX



each board's fingers pass through the other

29



### Through Finger (Box) Joint

Interlocking square fingers cut on the ends of both boards.

Moderate

#### FINGER & BOX



one outside face stays clean — nothing shows

30



### Half-Blind Box Joint

A finger joint whose fingers stop short of the far face, so one board's outside face stays perfectly flat and unbroken.

Hard

#### FINGER & BOX



looks like a mitre, holds like a box joint

31

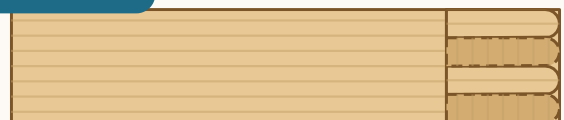


### Mitred Box Joint

A finger joint combined with a mitred outer corner.

Expert

#### FINGER & BOX



rounded finger ends

rounded fingers can be left proud — or let a lid pivot

32

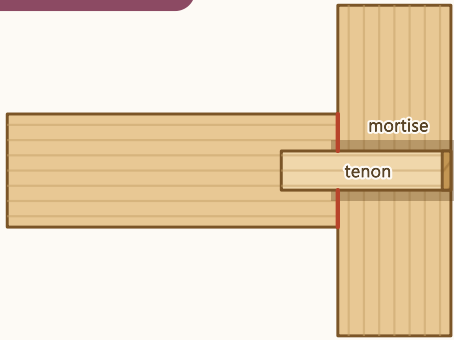


### Knuckle (Rounded Finger) Joint

A finger joint whose fingers are rounded off instead of squared.

Moderate

#### MORTISE & TENON



through tenon — its end grain shows on the far face

33

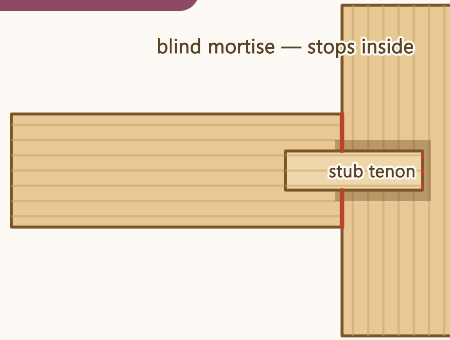


### Through Mortise & Tenon

The tenon is cut on the end of one piece and passes right through a mortise in the other, so its end grain shows on th.

Hard

#### MORTISE & TENON



nothing shows on the outside face

34

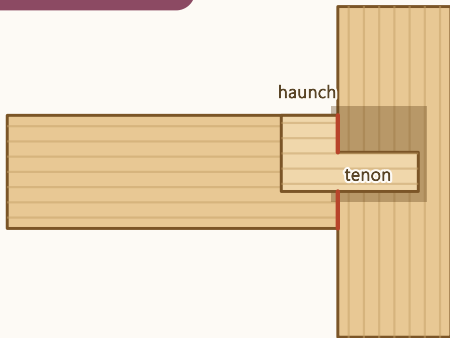


### Blind (Stub) Mortise & Tenon

A mortise that stops before it reaches the far face, with a matching stub tenon.

Hard

#### MORTISE & TENON



the haunch fills the panel groove and stops the rail twisting

35

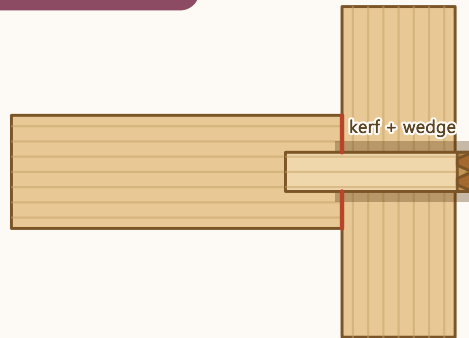


### Haunched Mortise & Tenon

A tenon with a short full-thickness extension at its top edge, called the haunch.

Hard

#### MORTISE & TENON



wedges splay the tenon and lock it — glue is almost optional

36

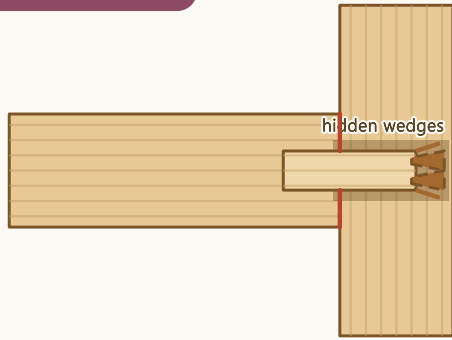


### Wedged Through Mortise & Tenon

A through tenon with saw kerfs cut in its end and wedges driven into them.

Expert

#### MORTISE & TENON



wedges are hidden inside — nothing shows on the outside

37

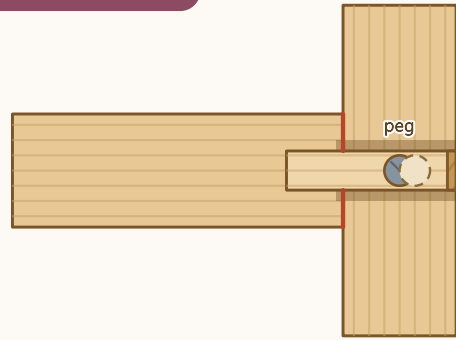


### Fox-Wedged (Blind Wedged) Tenon

A blind tenon that is wedged inside the mortise.

Expert

#### MORTISE & TENON



offset peg hole pulls the shoulder hard home

38

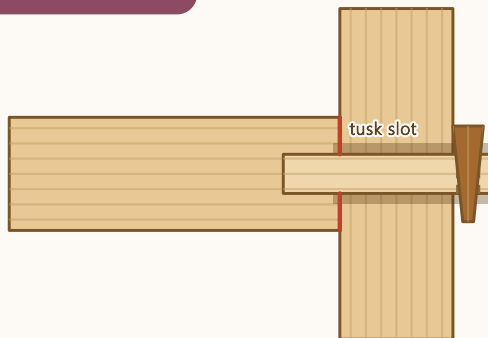


### Drawbored Mortise & Tenon

A through tenon with a peg driven through it.

Expert

#### MORTISE & TENON



knock the tusk out and the frame comes apart

39

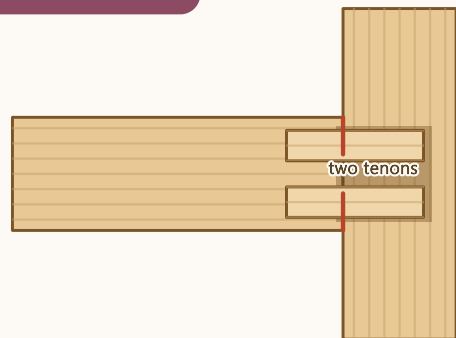


### Tusk Tenon (Knock-Down)

A through tenon that projects past the leg with a slot through it; a wedge-shaped tusk is driven through the slot from.

Hard

#### MORTISE & TENON



two tenons resist twisting far better than one wide tenon

40

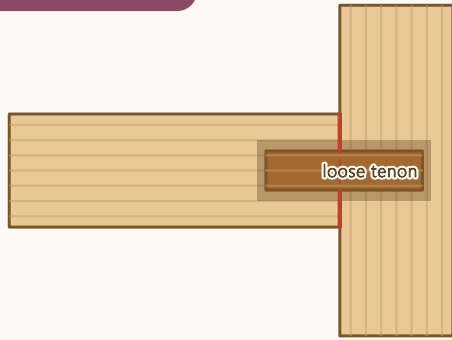


### Twin (Double) Mortise & Tenon

Two smaller tenons side by side instead of one wide one.

Hard

## MORTISE & TENON



a separate piece of wood bridges both mortises

41

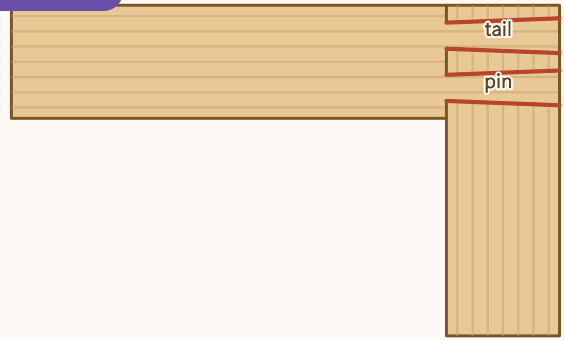


### Floating Tenon (Domino / Loose Tenon)

Neither piece has a tenon.

Easy

## DOVETAILS



tails pass right through — nothing can pull this apart sideways

42

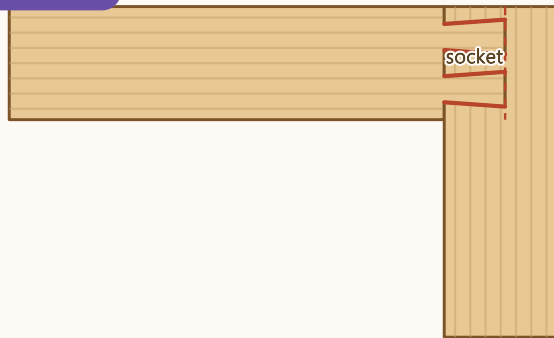


### Through Dovetail

Wedge-shaped tails on one board interlock with pins on the other, and the tails pass right through so the pattern show.

Expert

## DOVETAILS



tails stop short — the outside face stays clean

43

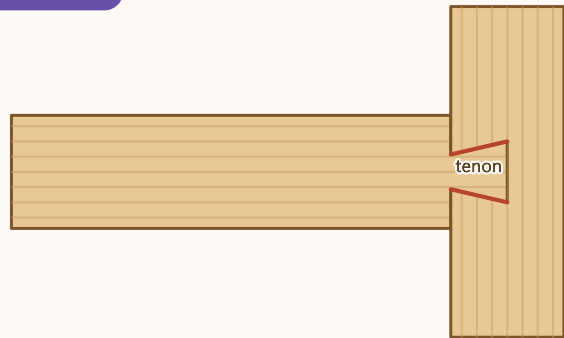


### Half-Blind Dovetail

The tails stop short of the far face, so the outside of the joint shows a clean flat board and only a thin line.

Expert

## DOVETAILS



slides in from the end — it cannot be pulled straight out

44

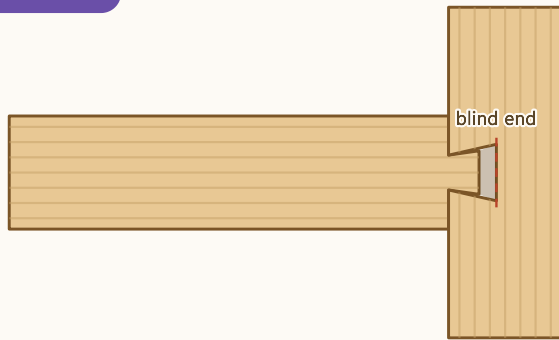


### Sliding Dovetail

A dovetail-shaped tenon that slides lengthwise into a matching dovetail groove.

Hard

## DOVETAILS



the groove stops short — the joint cannot slide through

45

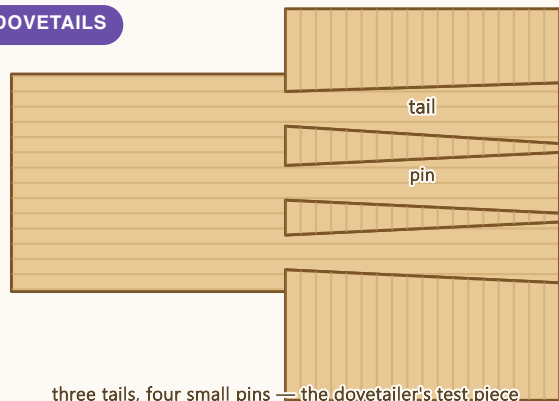


## Stopped Sliding Dovetail

A sliding dovetail whose groove is stopped before the far end, so the tenon runs in from one end and comes up against .

Expert

## DOVETAILS



three tails, four small pins — the dovetailer's test piece

46

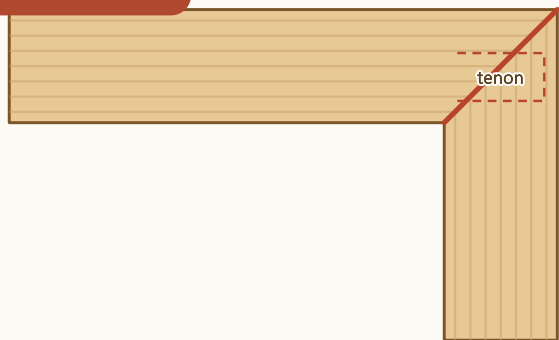


## Houndstooth Dovetail

A dovetail with small teeth instead of a full tail — three tails and four little pins in the space a normal dovetail u.

Expert

## CHINESE JOINERY



— mitres on every face, tenons hidden inside

47

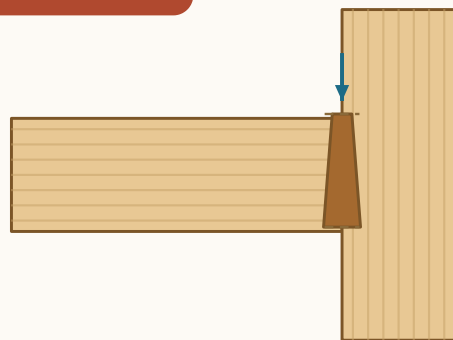


## Three-Way Mitred Corner

The Chinese rice-dumpling corner: a leg and two rails meeting at one corner, each face mitred so no end grain is visib.

Expert

## CHINESE JOINERY



— a dovetail key locks the joint and can be knocked out

48

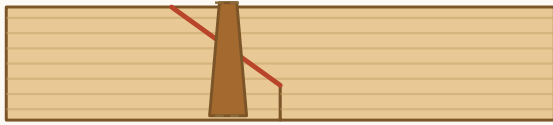


## Sliding Dovetail Key (Knock-Down)

A dovetail-shaped key driven into a matching slot across a joint.

Hard

## CHINESE JOINERY



— a driven wedge locks the scarf and hauls it tight

49

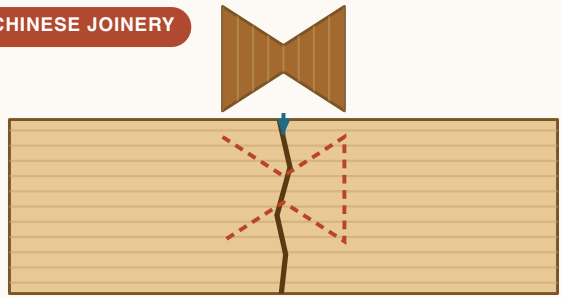


### Kanawa Tsugi (Japanese Scarf)

The Japanese 'metal-ring splice' — a long diagonal scarf with a wedge driven in behind it.

Expert

## CHINESE JOINERY



— bowtie key let into a slab to stop a crack running

50

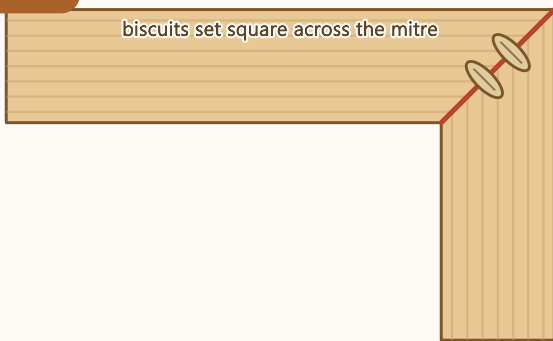


### Butterfly (Bowtie) Key

A bowtie-shaped inlay let into a board across a split.

Moderate

## MITRES



biscuits set square across the mitre

invisible from outside — glued in end grain both sides

51

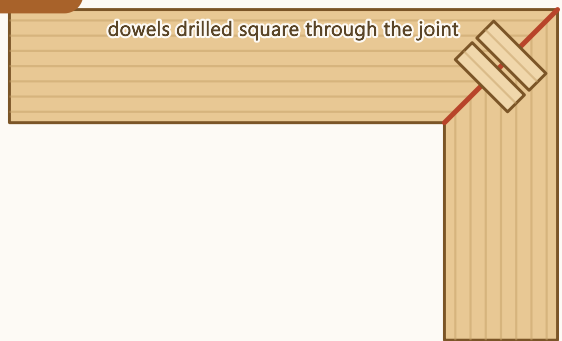


### Biscuited Mitre

A plain mitre reinforced with biscuits set across the joint.

Moderate

## MITRES



dowels drilled square through the joint

the simplest invisible reinforcement for a mitre

52



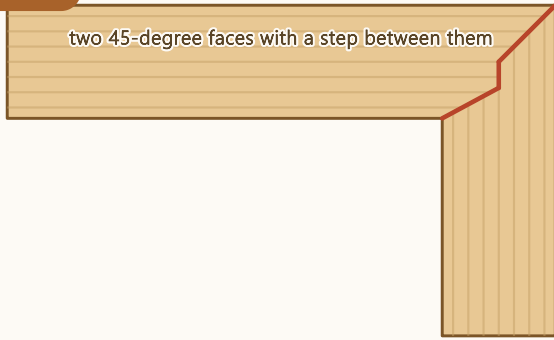
### Dowelled Mitre

A mitre reinforced with dowels drilled square through the joint.

Moderate

**MITRES**

two 45-degree faces with a step between them



the step locks it — it self-aligns as the clamps go on

53



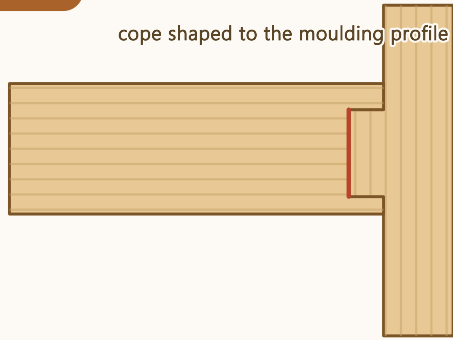
### Lock Mitre (Router Joint)

Two 45-degree faces with a square step between them, cut with a lock-mitre router bit.

Hard

**MITRES**

cope shaped to the moulding profile



internal corners do not open up the way a mitre does

54



### Coped Joint

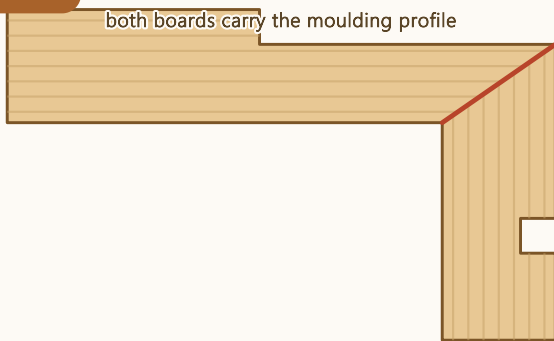
(Cope)

The joint for internal corners of moulding and skirting.

Hard

**MITRES**

both boards carry the moulding profile



compound mitre — the saw is tilted AND rotated

55



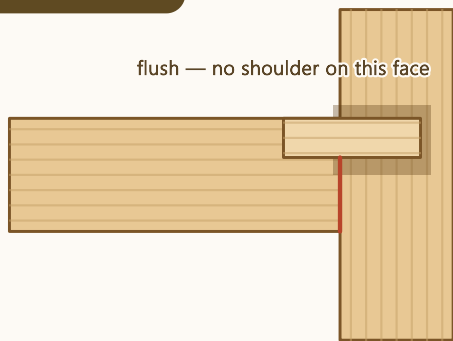
### Compound Mitre

A mitre where the saw is both tilted and rotated, so the joint follows two angles at once.

Hard

**TIMBER FRAMING**

flush — no shoulder on this face



one shoulder, flush outside — cheaper and weaker than a full tenon

56

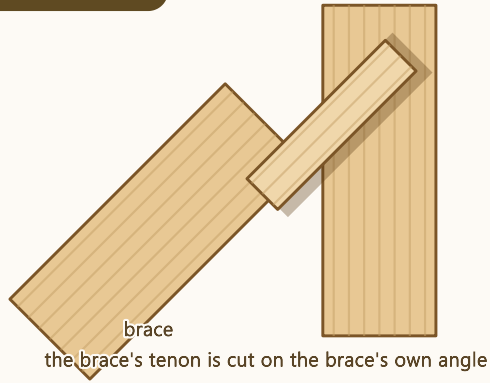


### Bare-Faced Tenon

A tenon reduced on one side only, so its outer face is flush with the rail's outer face.

Moderate

TIMBER FRAMING



57

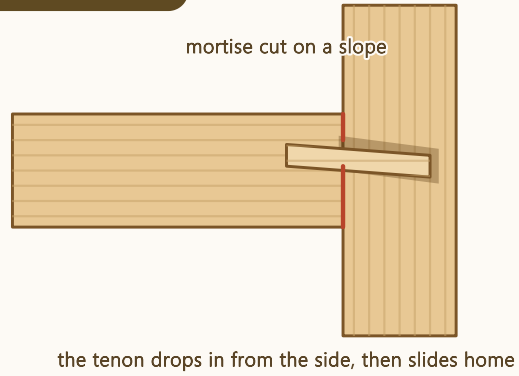


## Angled (Brace) Tenon

The joint at the end of a diagonal brace, cut on the same angle as the brace itself.

Hard

TIMBER FRAMING



58

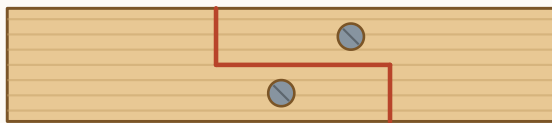


## Chase Mortise (Sloped Mortise)

A mortise cut on a slope instead of square, so the tenon can be dropped in from the side and then slid home.

Expert

TIMBER FRAMING



pegged through the lap — the standard length splice

59



## Half-Lap Scarf

A scarf made by halving both ends over a length and lapping them.

Moderate

TIMBER FRAMING



the cog stops the tie beam pulling away from the rafter

60



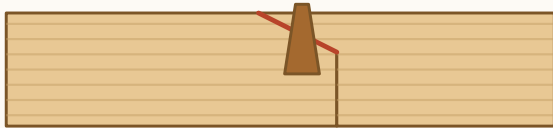
## Cogged Joint

A notch with a rectangular cog set in it, so the joint resists being pulled apart as well as being pushed together.

Expert

## TIMBER FRAMING

half-dovetail — one sloped face



the wedge hauls it tight and stops the creep

61



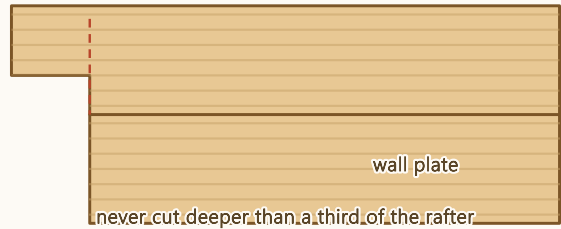
### Wedged Half-Dovetail

A rafter-to-plate joint where one face is cut on the dovetail slope and a wedge is driven in behind it.

Hard

## TIMBER FRAMING

rafter



wall plate

never cut deeper than a third of the rafter

62



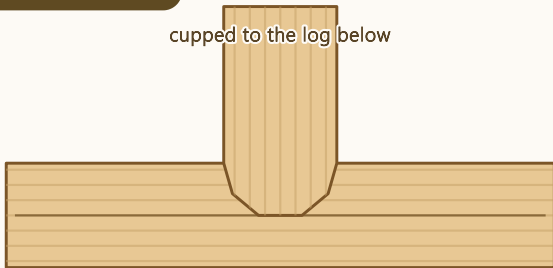
### Birdsmouth (Rafter Notch)

The right-angled notch cut in a rafter so it sits flat and square on the wall plate.

Easy

## TIMBER FRAMING

cupped to the log below



saddle notch — the wall settles without opening up

63



### Saddle Notch

The log-building joint: the upper log is cupped out on its underside so it saddles over the log below.

Moderate

## TIMBER FRAMING

taper on one face only



it drops straight on, then cannot lift off

64



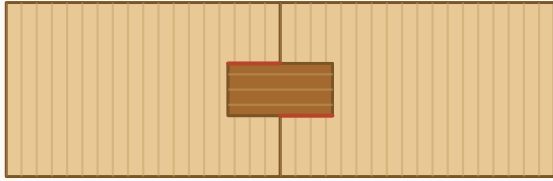
### Half-Dovetail Notch

A log notch with the dovetail taper cut on one face, so the upper log can be dropped straight down over the lower one

Hard

PANEL & EDGE

grooves in both edges



a separate strip does the work — neither board is cut

65

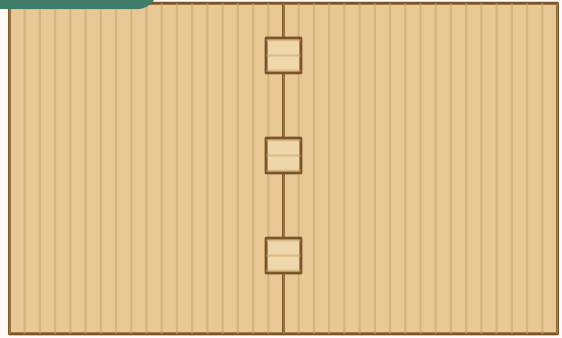


## Loose Tongue (Spline) Edge Joint

Two boards edge-jointed with a separate tongue — a loose spline — sitting in a groove cut in both edges.

Moderate

PANEL & EDGE



dowels keep the boards level while the glue sets

66

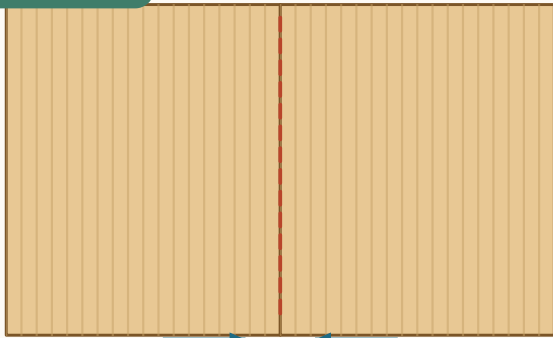


## Dowelled Edge Joint

Boards edge-jointed with dowels across the glue line.

Easy

PANEL & EDGE



rub until the boards suck together — glue only, no fasteners

67

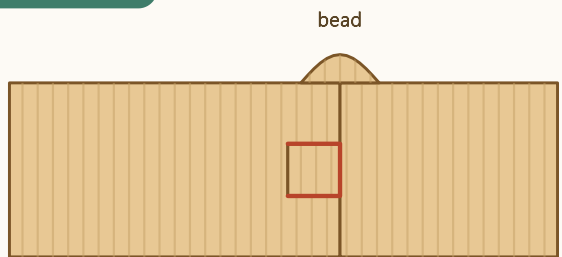


## Rub Joint (Glue-Only Edge)

Two boards joined with glue and nothing else.

Easy

PANEL & EDGE



the bead hides the seam and becomes a feature

68

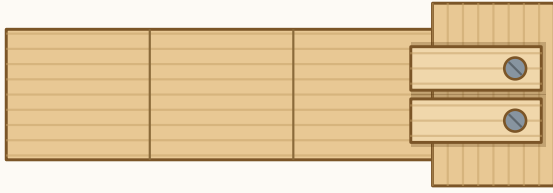


## Beaded Tongue & Groove

Tongue & groove with a bead worked on the face along one edge.

Moderate

PANEL & EDGE



top boards

the batten holds the top flat but lets it move

69

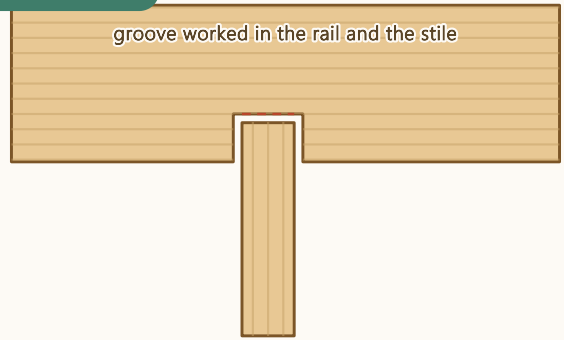


## Breadboard End

A cross-grain batten tenoned onto the ends of a tabletop.

Hard

PANEL & EDGE



the panel floats — never glue it along its length

70

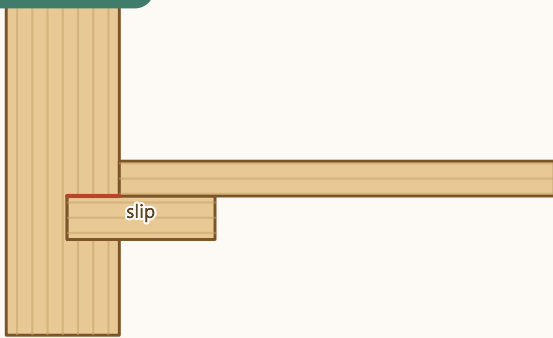


## Frame & Panel (Floating Panel)

A panel captured in grooves worked into a frame of rails and stiles.

Hard

PANEL & EDGE



the slip carries the bottom — the side keeps its strength

71

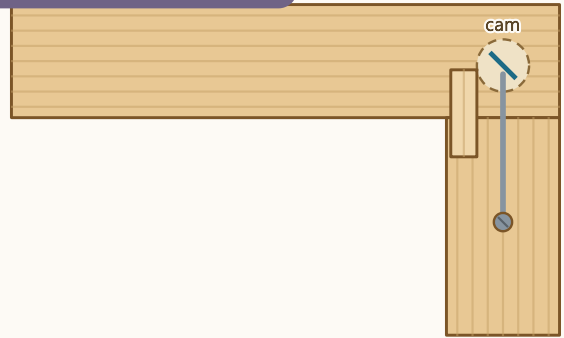


## Drawer Slip (Runner) Joint

A thin strip let into a groove in the drawer side, projecting inward to carry the drawer bottom.

Moderate

KNOCK-DOWN & HARDWARE



a quarter turn of the cam draws the panels together

72

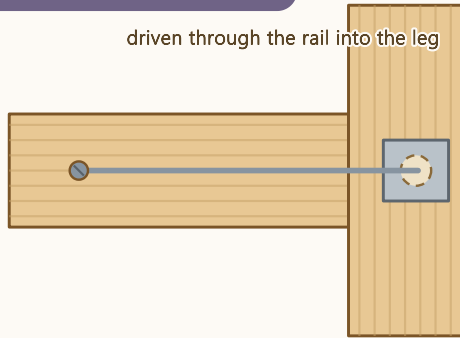


## Cam & Dowel (Minifix)

The flat-pack connector: a dowel bolt screwed into one panel engages a rotary cam in the other, and a quarter turn of .

Easy

#### KNOCK-DOWN & HARDWARE



bed bolts tighten to a rattle-free fit and still come apart

73

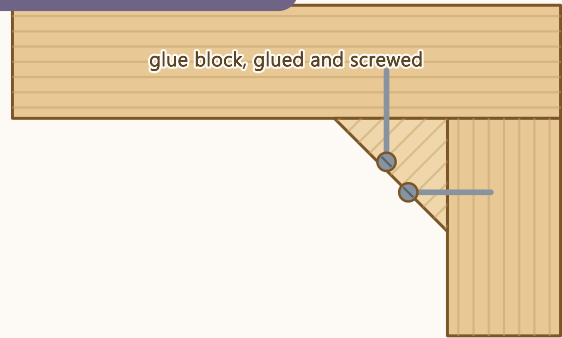


### Bed Bolt Joint

A heavy bolt driven through the bed rail into a cross-dowel nut set in the leg.

Moderate

#### KNOCK-DOWN & HARDWARE



squares the joint and takes the racking

74

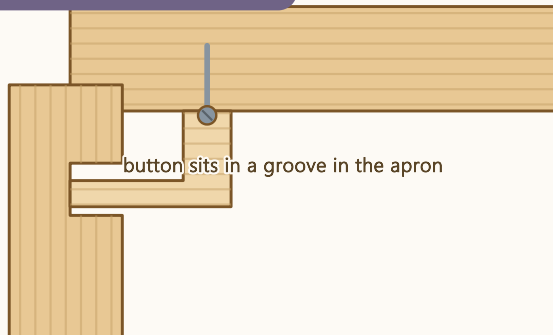


### Corner Block (Glue Block)

A triangular block glued and screwed into an inside corner.

Easy

#### KNOCK-DOWN & HARDWARE



holds the top down but lets it slide as it moves

75

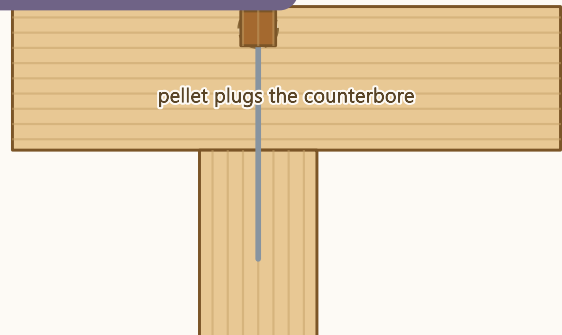


### Table Button (Top Fastener)

A small tongue screwed into a groove in the apron and to the underside of the top.

Moderate

#### KNOCK-DOWN & HARDWARE



the quick fixing that finishes like joinery

76



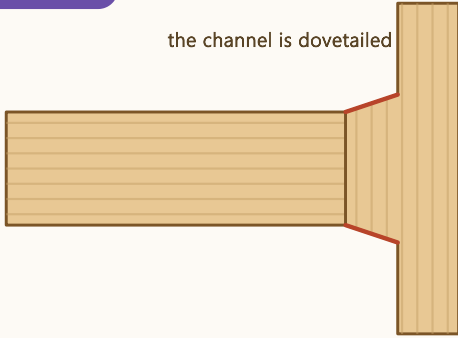
### Screw & Pellet

A countersunk screw with the head buried below the surface under a glued-in wooden plug cut from the same board.

Easy

## DOVETAILS

the channel is dovetailed



the shelf slides in and then cannot be pulled out

77



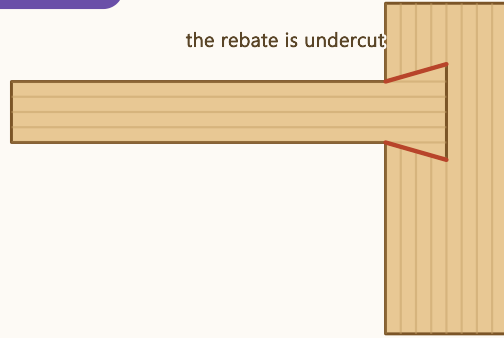
## Dovetail Dado (Dovetail Housing)

A shelf housed into a dovetail channel cut across a carcass side.

Hard

## DOVETAILS

the rebate is undercut



the rebate locates it, the dovetail locks it

78



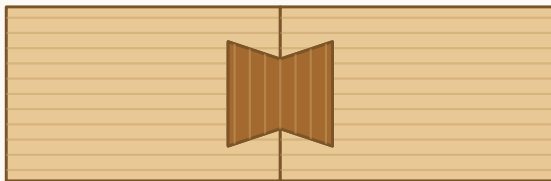
## Dovetailed Rabbet

A rebate whose long-grain face is undercut into a dovetail, so the mating board locks as it seats.

Expert

## DOVETAILS

the key is dovetailed at both ends



it cannot slide out either way

79



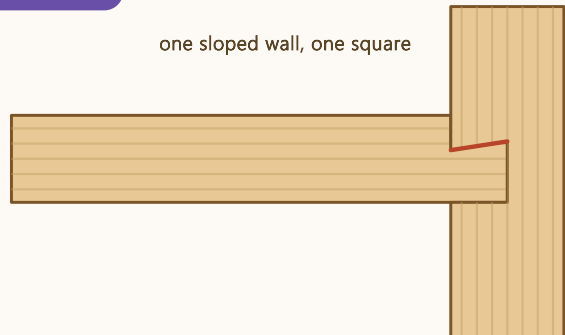
## Dovetail Key (Loose Dovetail)

Two grooves cut in the mating edges, dovetailed at the base, with a separate key that is dovetailed at BOTH ends.

Hard

## DOVETAILS

one sloped wall, one square



half the cutter work, most of the holding

80

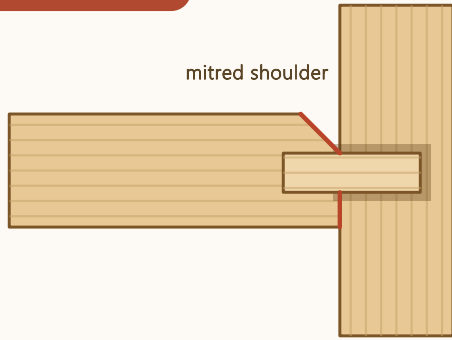


## Half-Dovetail Housing

A sliding dovetail with only one sloped face.

Moderate

CHINESE JOINERY



mitred shoulder

— a mitred shoulder instead of a square one

81

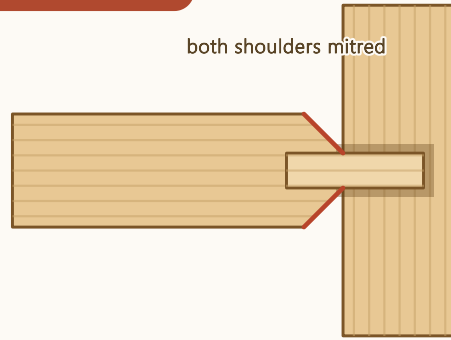


## Shouldered Mitred Tenon

— the tenon's shoulder is cut as a mitre rather than a square, so the rail meets the leg's corner without showing.

Expert

CHINESE JOINERY



both shoulders mitred

— the leg runs on past the apron

82

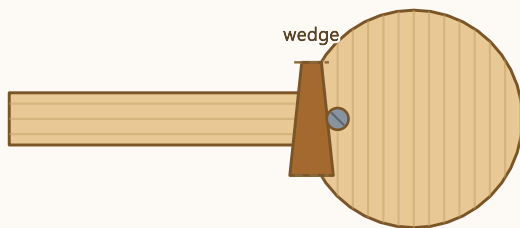


## Mitred Shouldered Joint

— the leg runs on past the apron and both of the apron's shoulders are mitred, so the shoulder wraps around the leg.

Expert

CHINESE JOINERY



wedge

— a wedge draws it in, a pin keeps it there

83

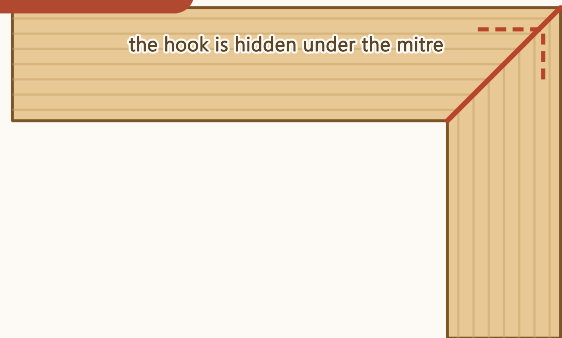


## Round Member Wedge & Pin

— the joint for round members.

Hard

CHINESE JOINERY



the hook is hidden under the mitre

— a hooked tenon inside a mitred frame corner

84



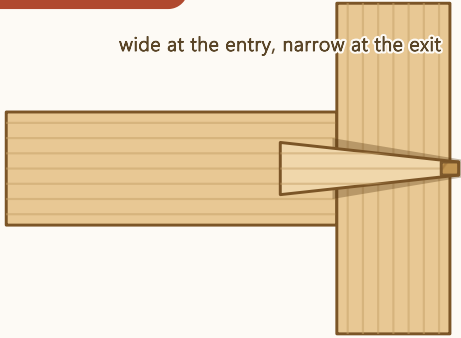
## Hooked Frame Corner

— the opium-pipe corner.

Expert

## CHINESE JOINERY

wide at the entry, narrow at the exit



— easy to assemble, then pinned

85



### Wide-In Narrow-Out Tenon

— a through tenon that is wide where it enters the mortise and narrow where it exits.

Hard

## JAPANESE JOINERY



ぎ — seated on a step, locked with a wedge

86



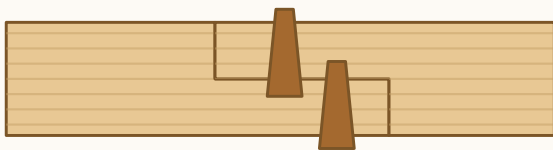
### Seated Scarf

ぎ

ぎ — a scarf where the two ends seat into each other on a step and a wedge driven across the joint hauls them tight.

Hard

## JAPANESE JOINERY



け ぎ — two driven keys lock the splice

87



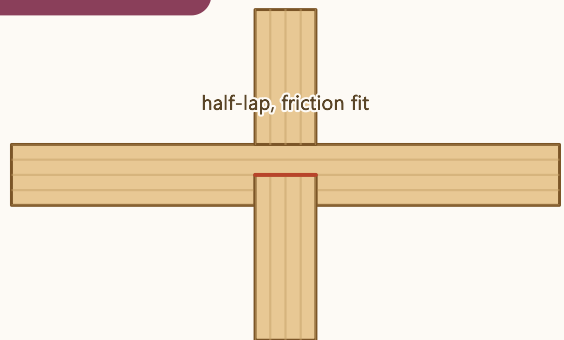
### Keyed Scarf Splice

け ぎ

け ぎ — a lapped scarf locked by two large wooden keys driven through it.

Expert

## JAPANESE JOINERY



half-lap, friction fit

— no glue anywhere, the fit does everything

88



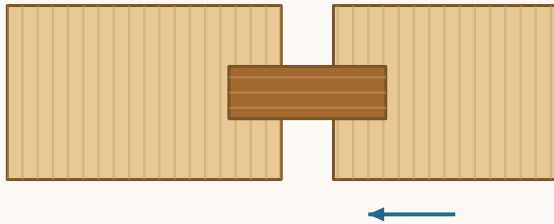
### Kumiko Lattice Joint

— the lattice joint.

Expert

## JAPANESE JOINERY

grooves in both edges



い — a separate tongue does the work

89



### Loose Tongue (Yatoi Zane)

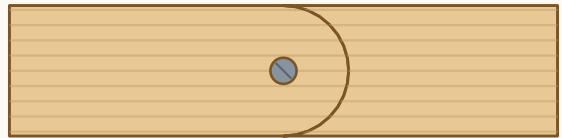
い

い — a loose tongue employed to join two boards.

Moderate

## SPECIALITY

the hinge pin sits on the arc centre



down it is a bullnose, up it is a flat top

90



### Rule Joint (Drop-Leaf)

The joint that makes a drop-leaf table work.

Expert

## FINGER & BOX



for joining lengths, not corners — stronger than the wood

91

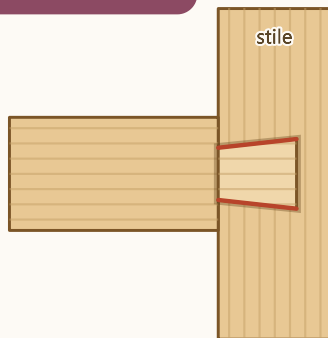


### End-to-End Finger Joint

The finger joint used for its other purpose: joining boards END TO END to make a longer one.

Moderate

## MORTISE & TENON



the dovetail section means it can only slide in from one end

92

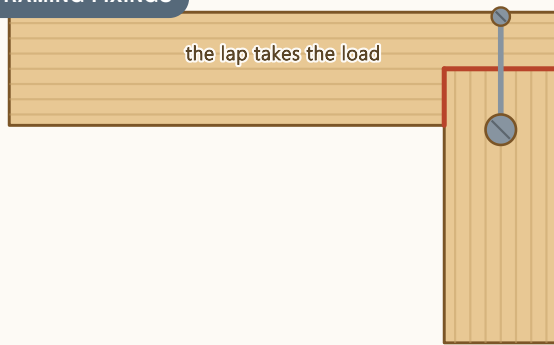


### Dovetail Tenon

A tenon whose cross-section is a dovetail, so once it is in the mortise it cannot be pulled straight out — only slid i.

Expert

FRAMING FIXINGS



bolts and washers do the work — always use a washer

93

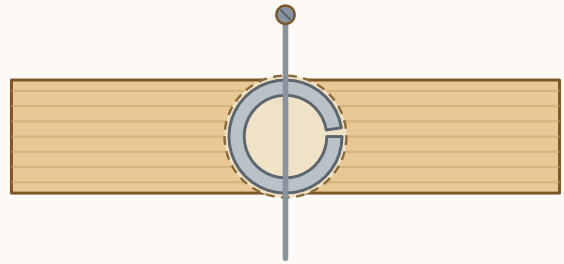


## Bolted Lap Joint

A half-lap drawn up tight with a bolt and washer.

Easy

FRAMING FIXINGS



a split ring multiplies the joint's load capacity

94

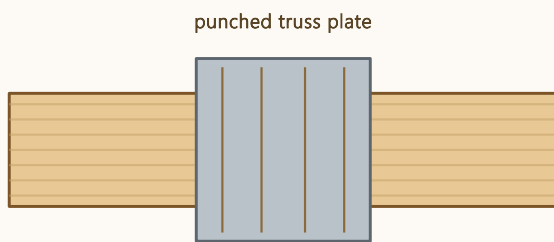


## Split-Ring Connector

A split metal ring let into matching circular grooves in two timber faces.

Moderate

FRAMING FIXINGS



the teeth carry the load in shear — pressed, never hammered

95

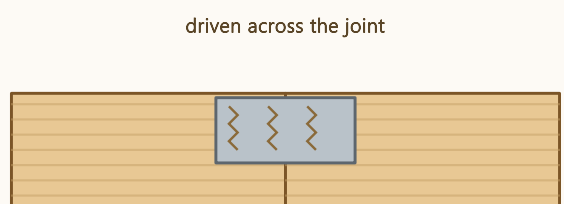


## Punched Metal Truss Plate

A galvanised steel plate punched with teeth, pressed into both faces of a joint.

Easy

FRAMING FIXINGS



fast and cheap — the fastener you never see on cheap furniture

96



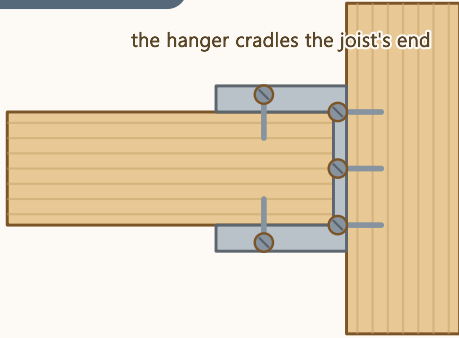
## Corrugated Fastener

A short wavy steel strip driven across a joint on the back face.

Easy

## FRAMING FIXINGS

the hanger cradles the joist's end



every nail works in shear — use the specified nails

97



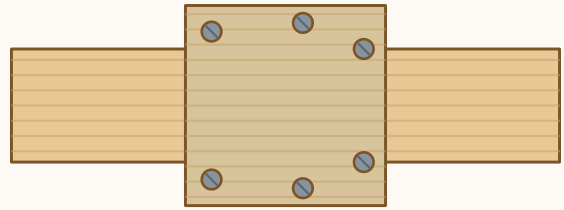
### Joist Hanger

A U-shaped galvanised hanger nailed to the beam, cradling the joist's end.

Easy

## FRAMING FIXINGS

the plate straddles the joint



a gusset spreads the load across the whole joint

98

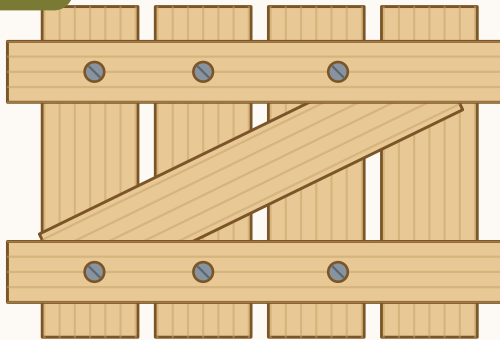


### Gusset (Mending) Plate

A flat plate screwed or glued over a joint to spread the load and repair a break.

Easy

## SPECIALITY



ledged & braced — the boards do nothing structural

99

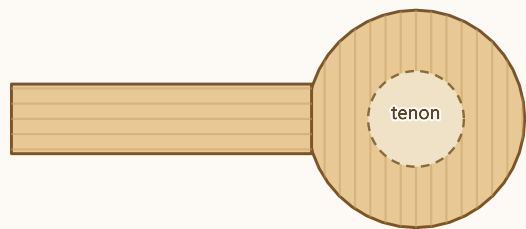


### Lugged & Braced (Batten) Joint

Boards nailed to horizontal ledges with a diagonal brace between them.

Easy

## MORTISE & TENON



round M&T — no shoulder, so the fit is everything

100



### Round Mortise & Tenon

A turned round tenon into a bored round mortise — no shoulders, no square work, and nothing to hide a bad fit.

Hard